

5.4.17

EE25BTECH11022 - sankeerthan

Question: Using elementary transformations, find the inverse of the following matrix.

$$\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$$

Solution: Given

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \quad (1)$$

Let $\mathbf{A}^{-1}$ be the inverse of $\mathbf{A}$. Then

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \quad (2)$$

Augmented matrix of $(\mathbf{A} \mid \mathbf{I})$ is given by

$$\left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 5 & 2 & 0 & 1 \end{array} \right) \xrightarrow{R_2 \rightarrow 3R_2 - 5R_1} \left(\begin{array}{cc|cc} 3 & 1 & 1 & 0 \\ 0 & 1 & -5 & 3 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 - R_2} \left(\begin{array}{cc|cc} 3 & 0 & 6 & -3 \\ 0 & 1 & -5 & 3 \end{array} \right) \quad (3)$$

$$\left(\begin{array}{cc|cc} 3 & 0 & 6 & -3 \\ 0 & 1 & -5 & 3 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -5 & 3 \end{array} \right) \quad (4)$$

Hence the inverse of the matrix $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ is $\begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$